

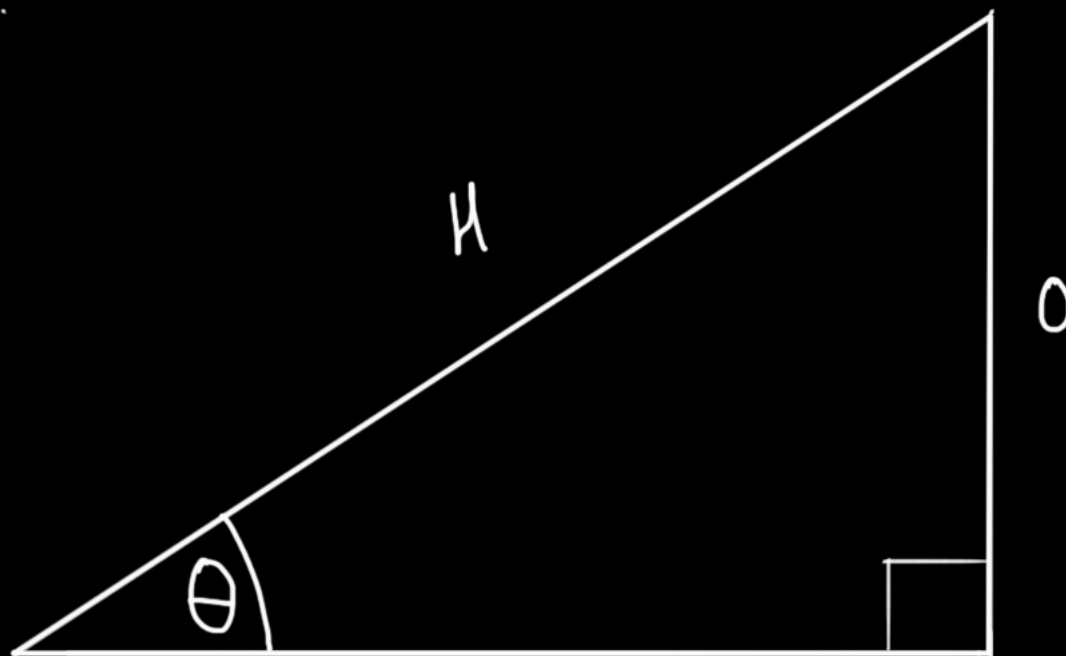
# TRIG SPECIAL SECTION

# REVIEW

- You are probably used to seeing trig problems being solved using SOH-CAH-TOA
  - SOA Sine = Opposite / Hypotenuse
  - CAH Cosine = Adjacent / Hypotenuse
  - Tangent = Opposite / Adjacent

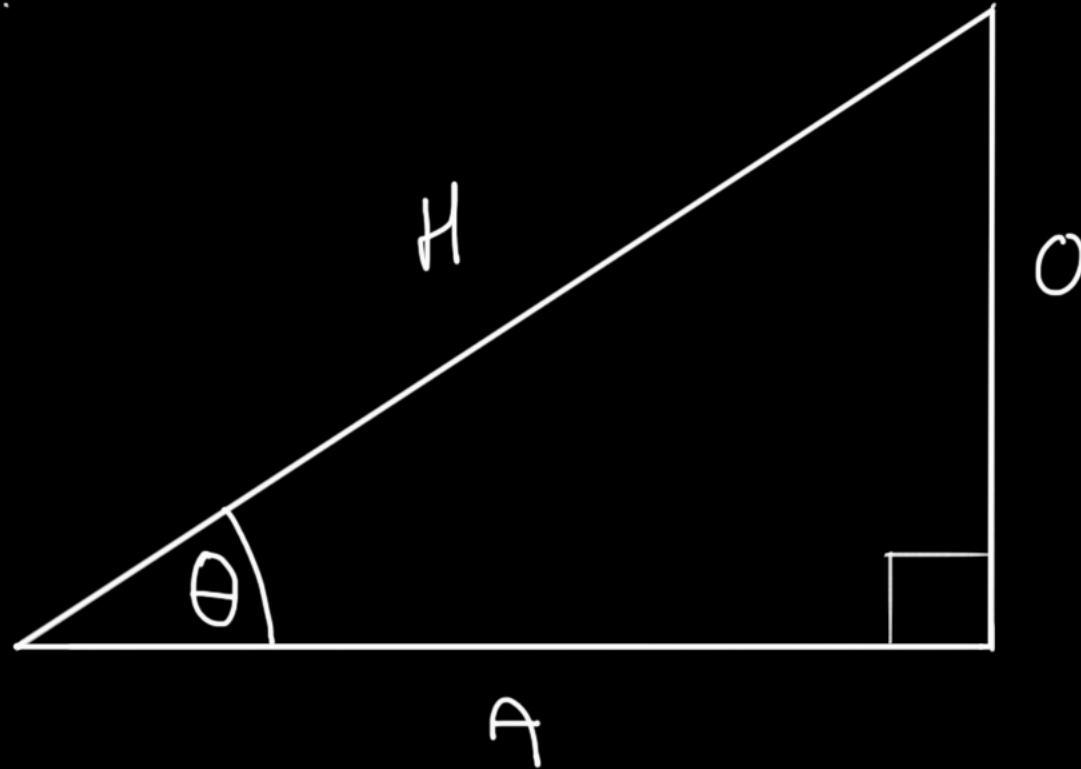
**SOH CAH TOA**

**SINE = OPPOSITE/HYPOTENUSE**



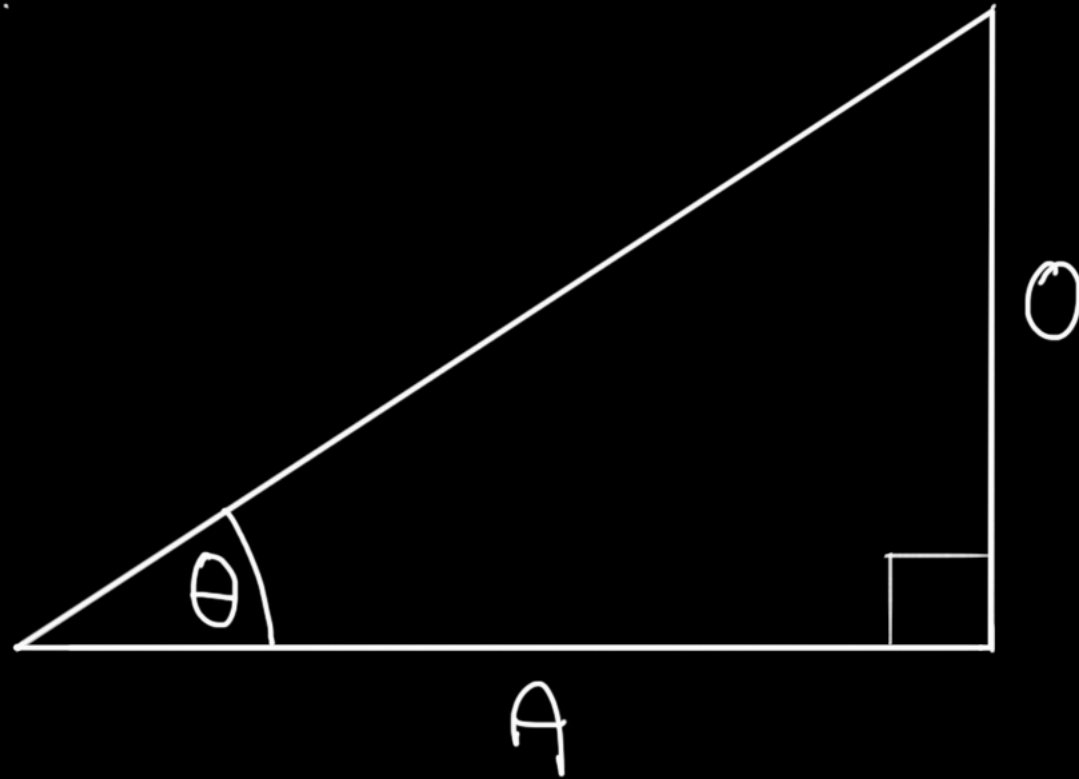
**SOH CAH TOA**

**COSINE = ADJACENT/HYPOTENUSE**



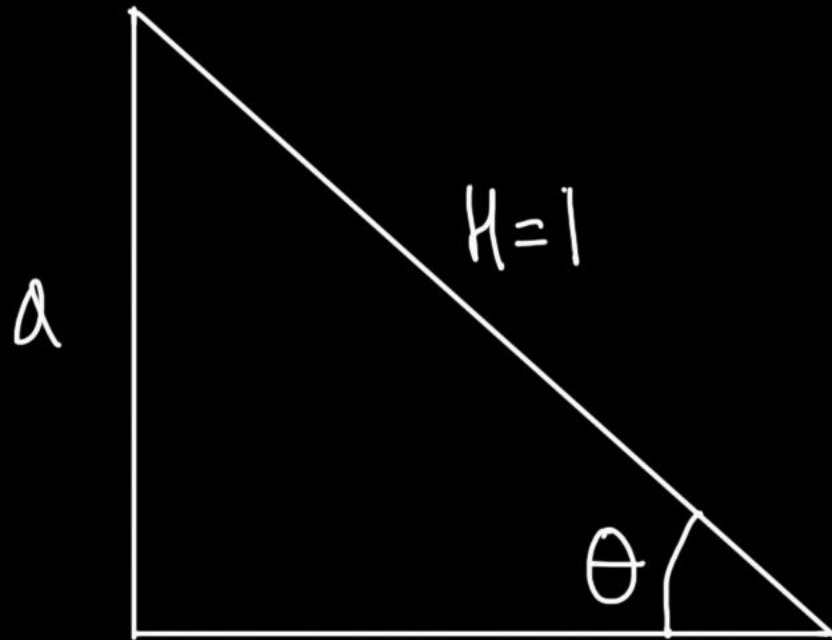
**SOH CAH TOA**

**TANGENT = OPPOSITE/ADJACENT**



# SOH CAH TOA

**SINE = OPPOSITE/HYPOTENUSE**



$$\theta = 45$$

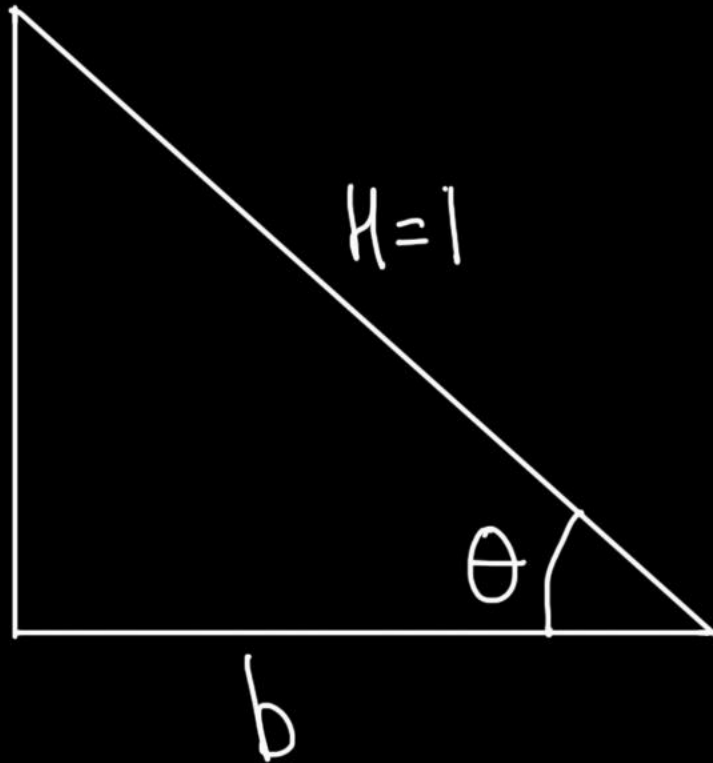
$$\sin(45) = 0.707$$

$$0.707 = \frac{a}{1}$$

$$a = 0.707$$

**SOH CAH TOA**

**COSINE = ADJACENT/HYPOTENUSE**



$$\theta = 45$$

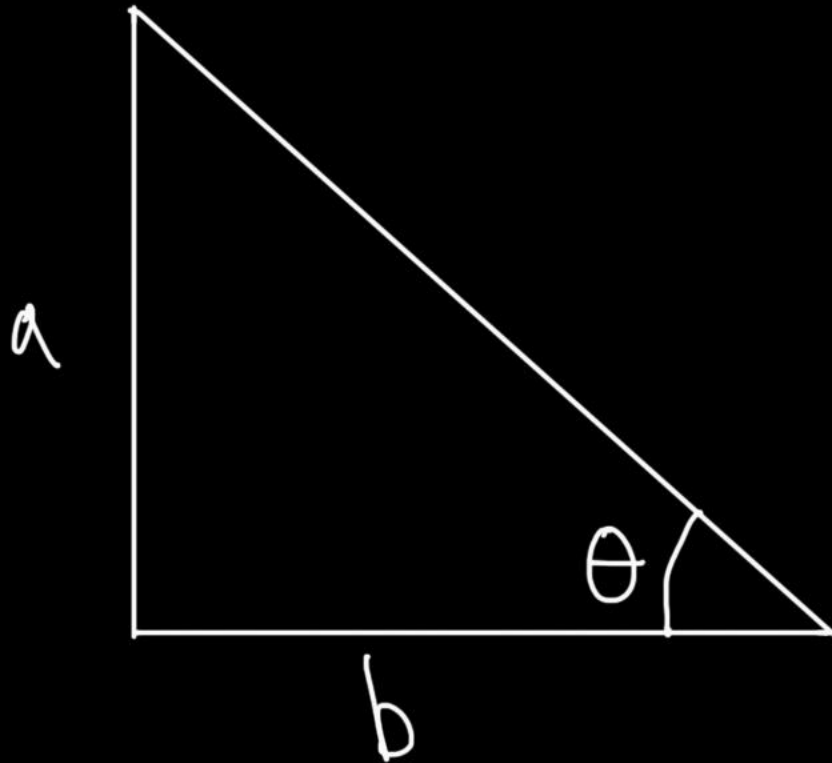
$$\cos(45) = 0.707$$

$$0.707 = \frac{b}{1}$$

$$b = 0.707$$

# SOH CAH TOA

**TANGENT = OPPOSITE/ADJACENT**



$$\theta = 45$$

$$\tan(45) = 1$$

$$1 = \frac{a}{b}$$

$$a = b$$

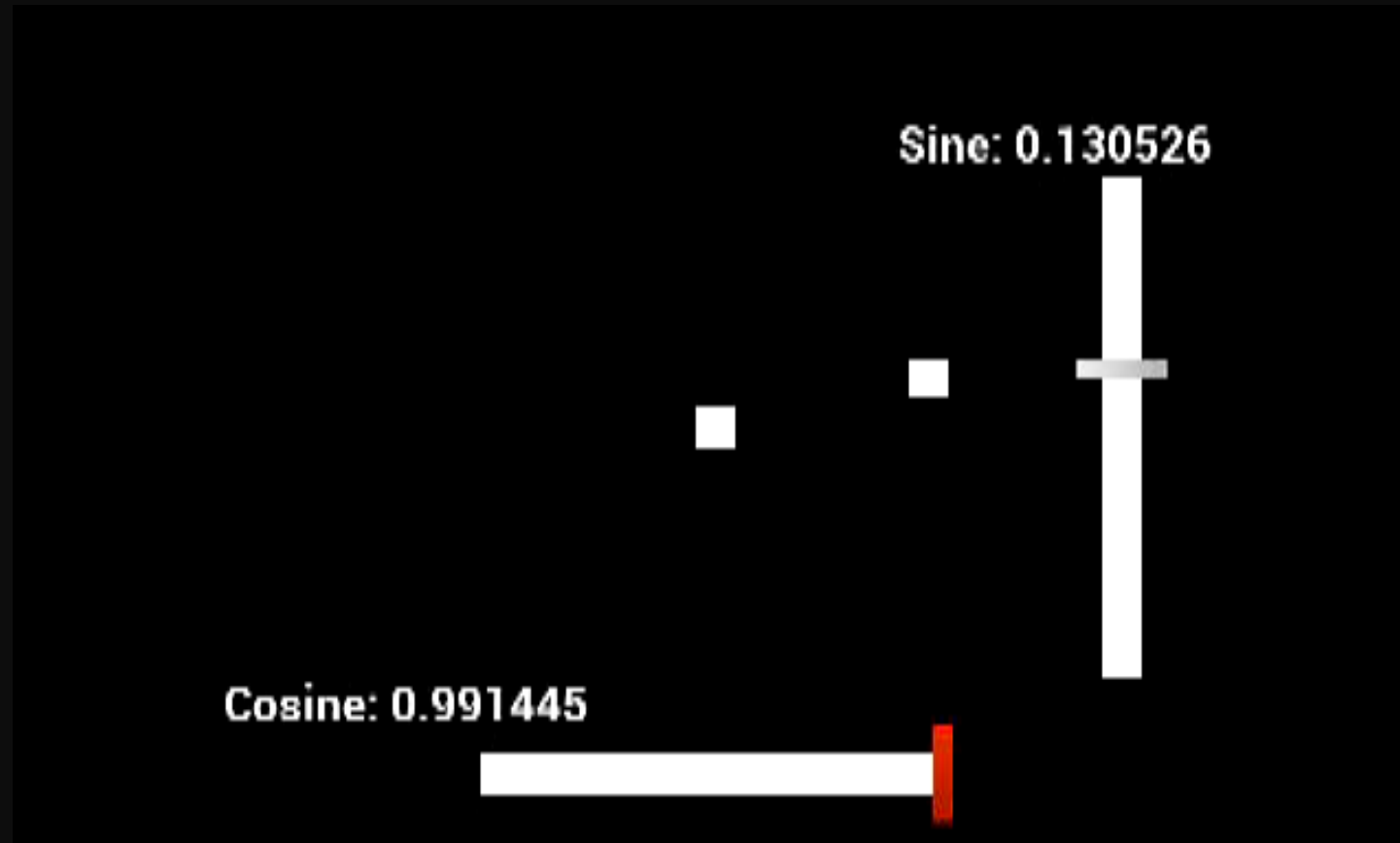


# TANGENT?

- As you can see from the previous example, the Tangent of an angle is the ratio of the length of the opposite side of a right angle triangle over the adjacent length.
- What this is really talking about is how much each contributes to the hypotenuse.
- While the Sine is (typically) the Y axis, and the Cosine is the X, the Tangent is the ratio between how much the Y and X axis each contribute to the hypotenuse.

# WHAT ARE SINE AND COSINE ANYWAYS?

(Play Video)→

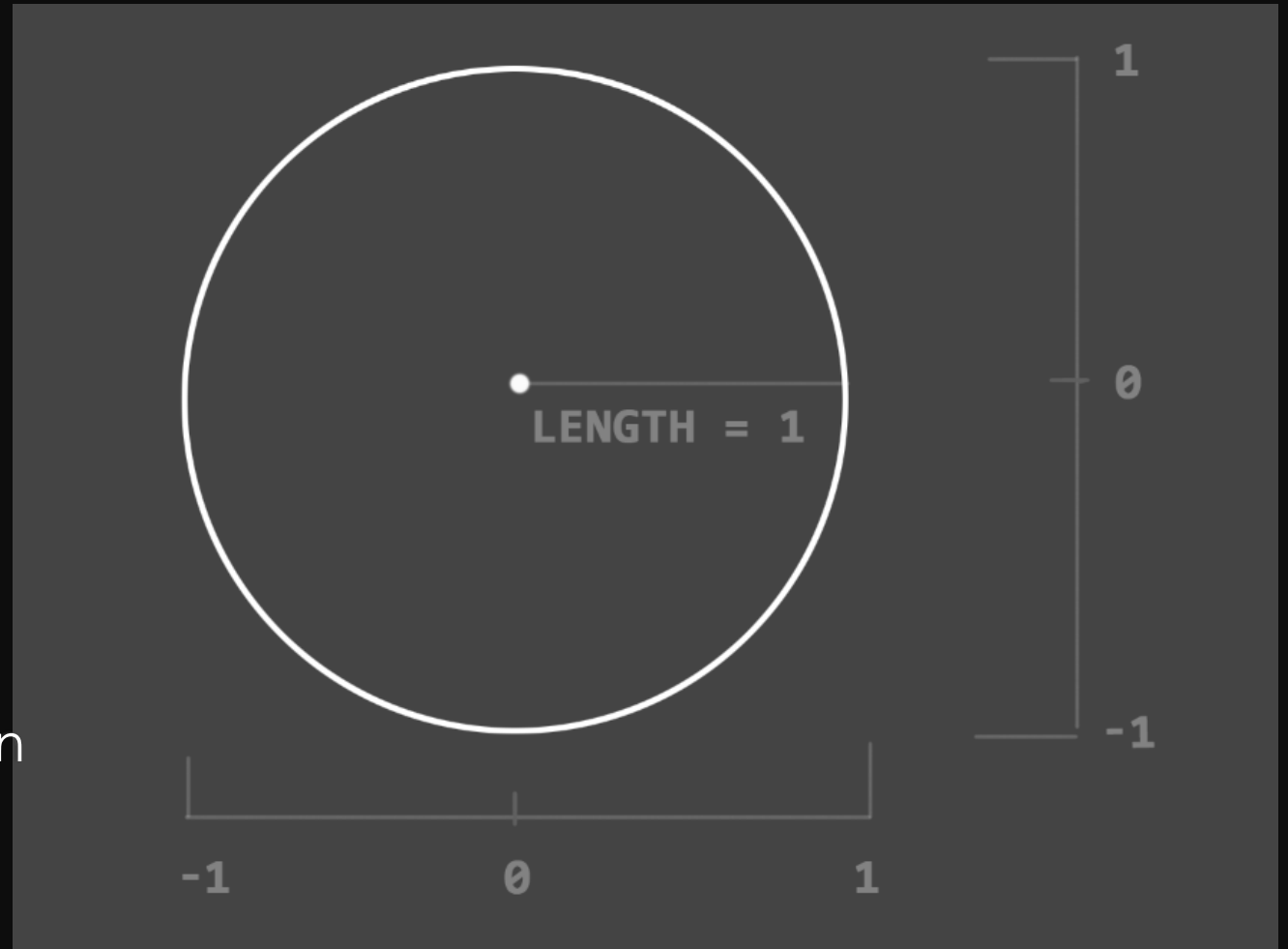


# WHAT IS GOING ON HERE?

- To understand what  $\sin$  and  $\cos$  really mean, we will look at the unit circle
- The unit circle is a circle with a radius of 1

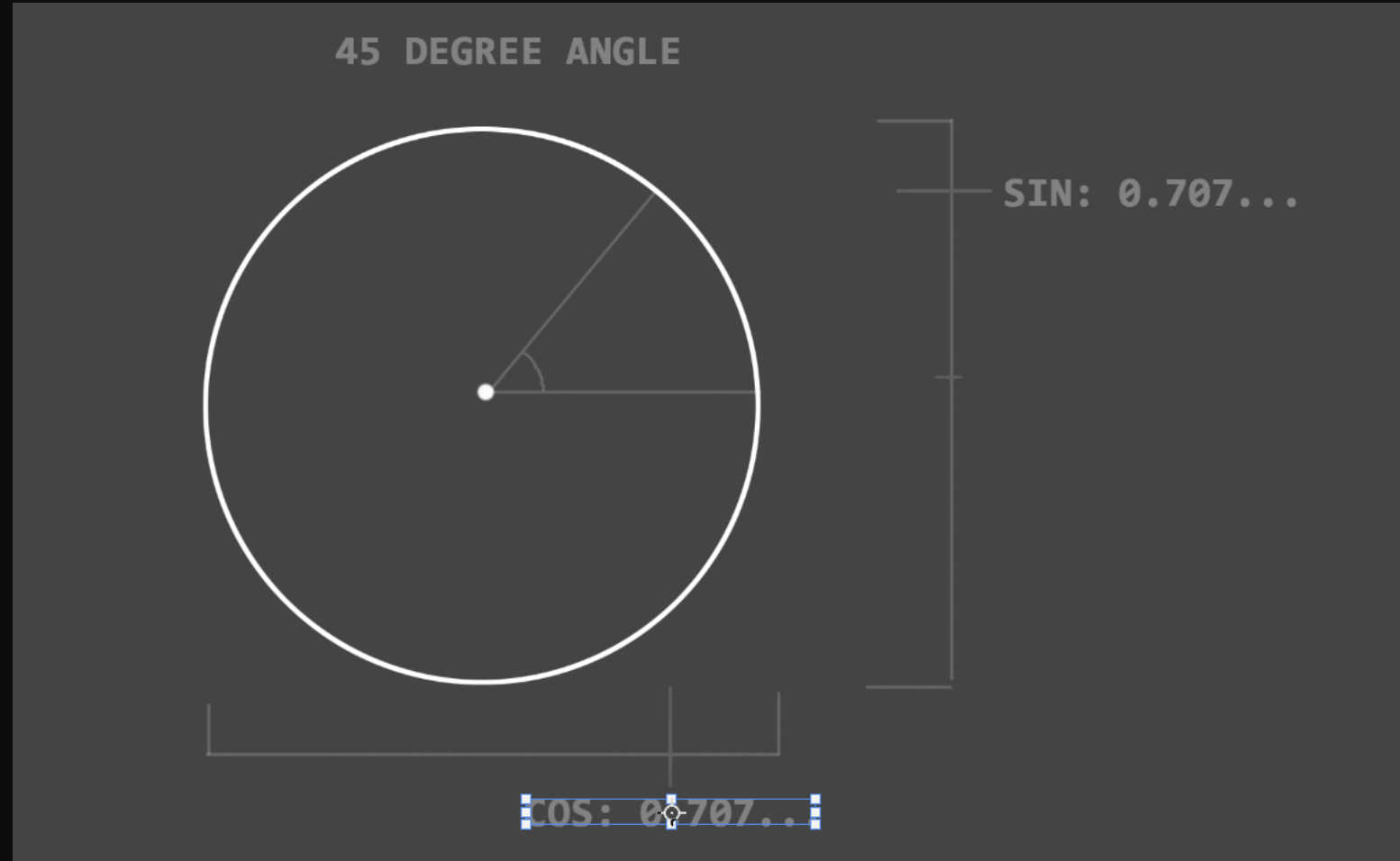
# UNIT CIRCLE

- Here we have a unit circle, including position coordinates
- (0,0) is the center point
- The line shown would go from (0,0) to (1,0)
- Sine represents the proportion of the position of a point on the unit circle on the Y axis
- Cosine represents the proportion of the position of a point on the unit circle on the X axis



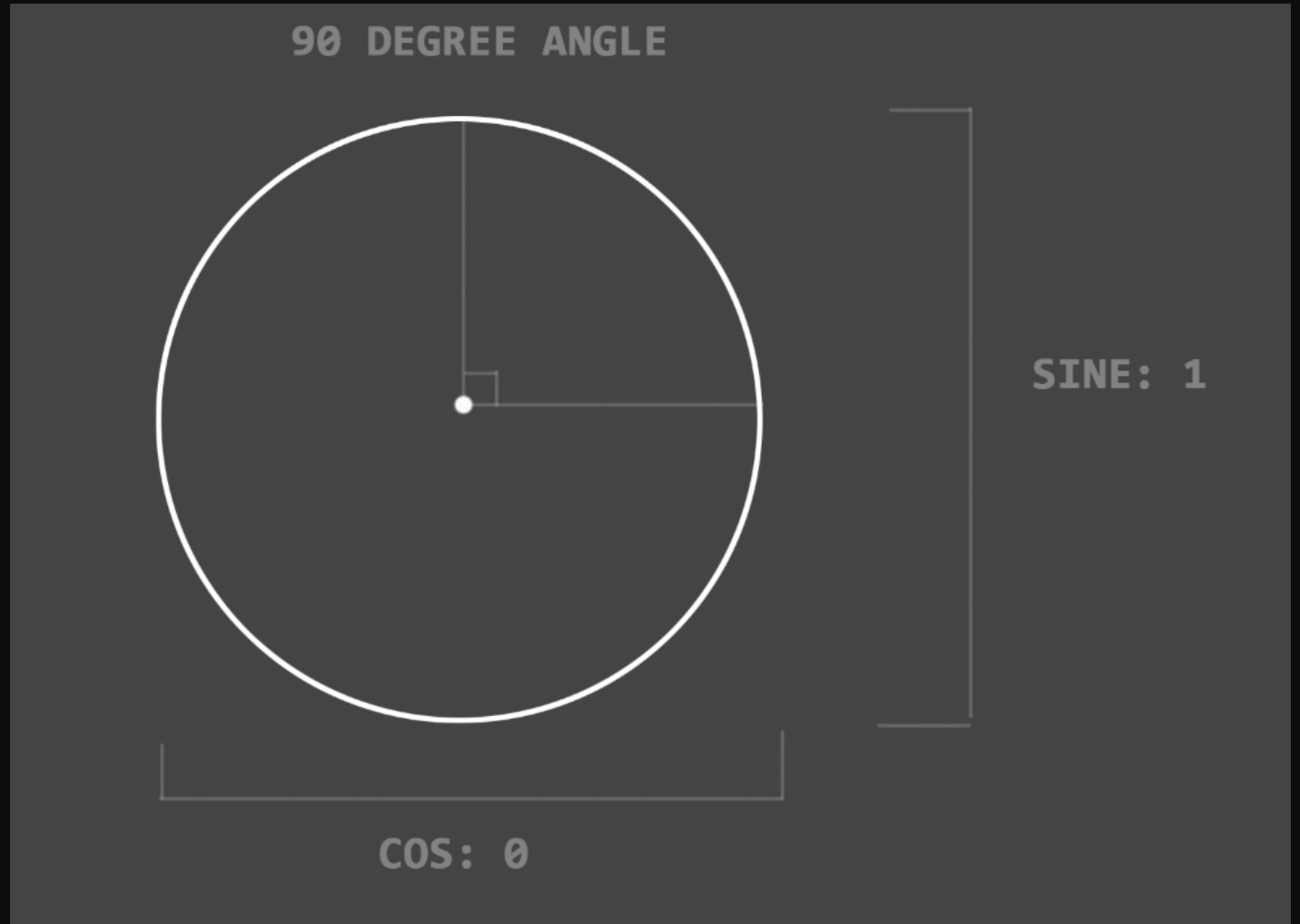
# UNIT CIRCLE

- If we take the sin and cos of a  $45^\circ$  angle we get:
- $\sin = 0.707$
- $\cos = 0.707$
- As you can see, this represents the position on the Y Axis (for Sin) and the position on the X Axis (for cos) of the line as it intersects the circle.



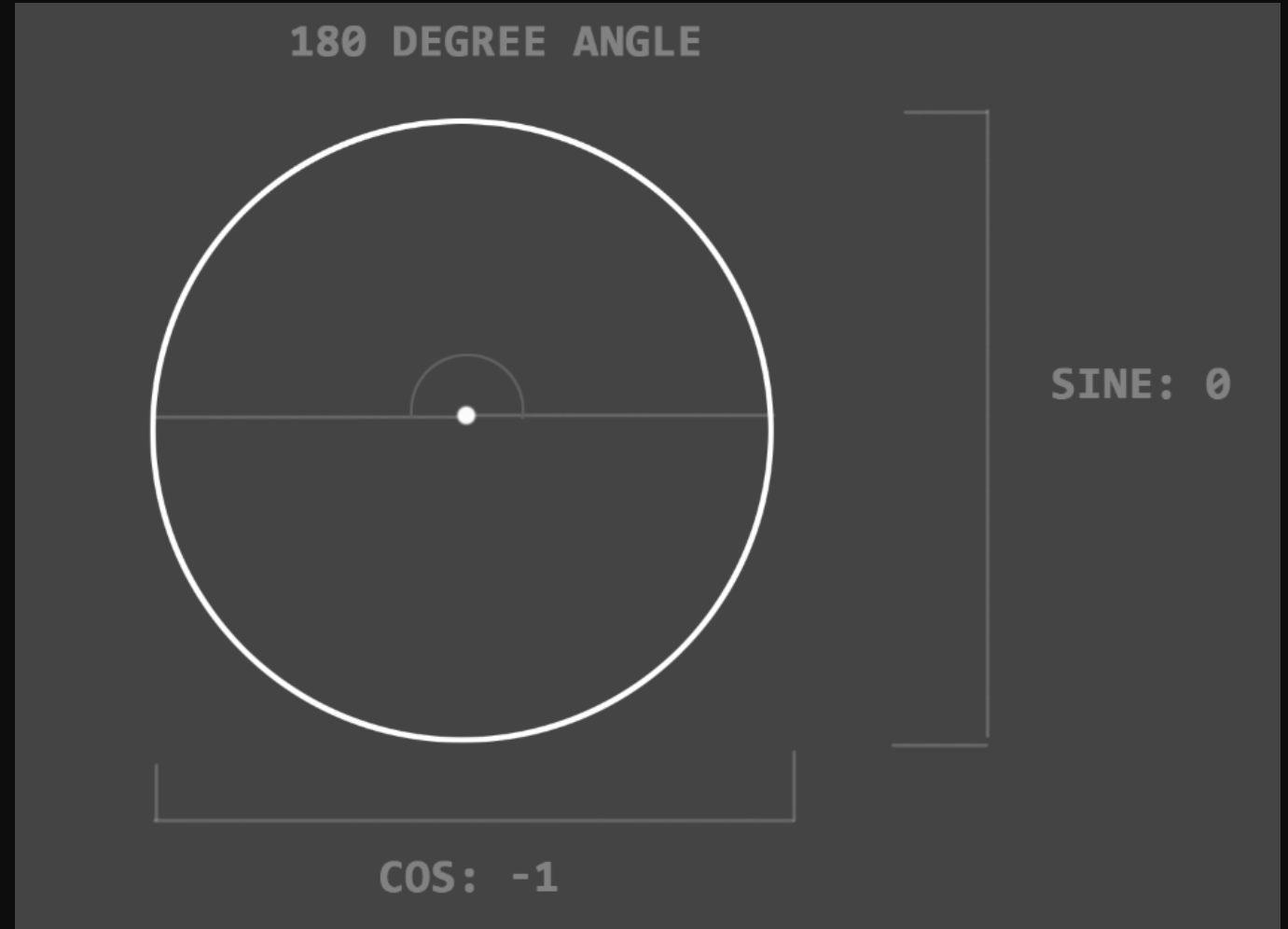
# UNIT CIRCLE

- Looking at some other examples, the same relationship holds.



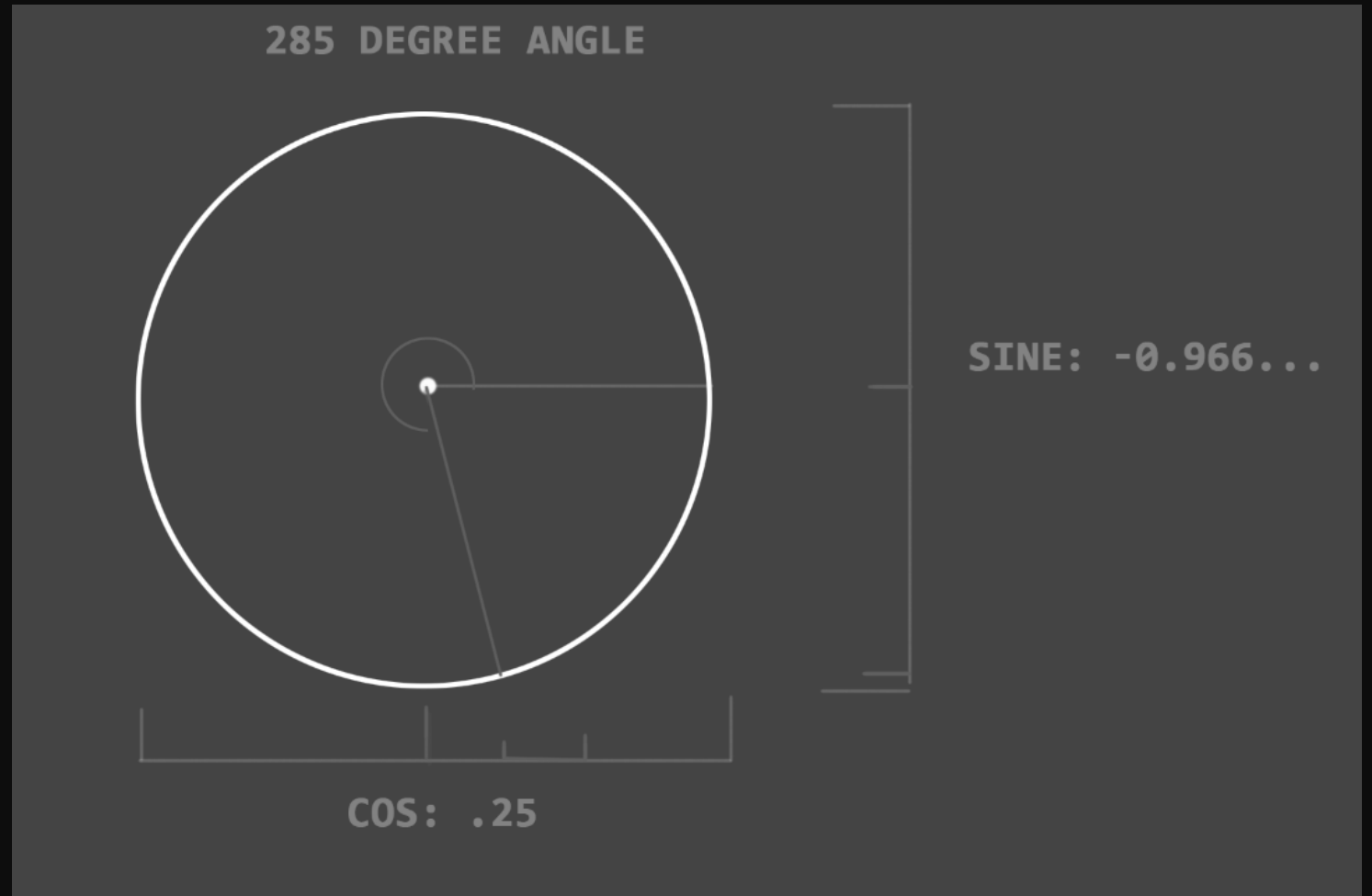
# UNIT CIRCLE

- Looking at some other examples, the same relationship holds.



# UNIT CIRCLE

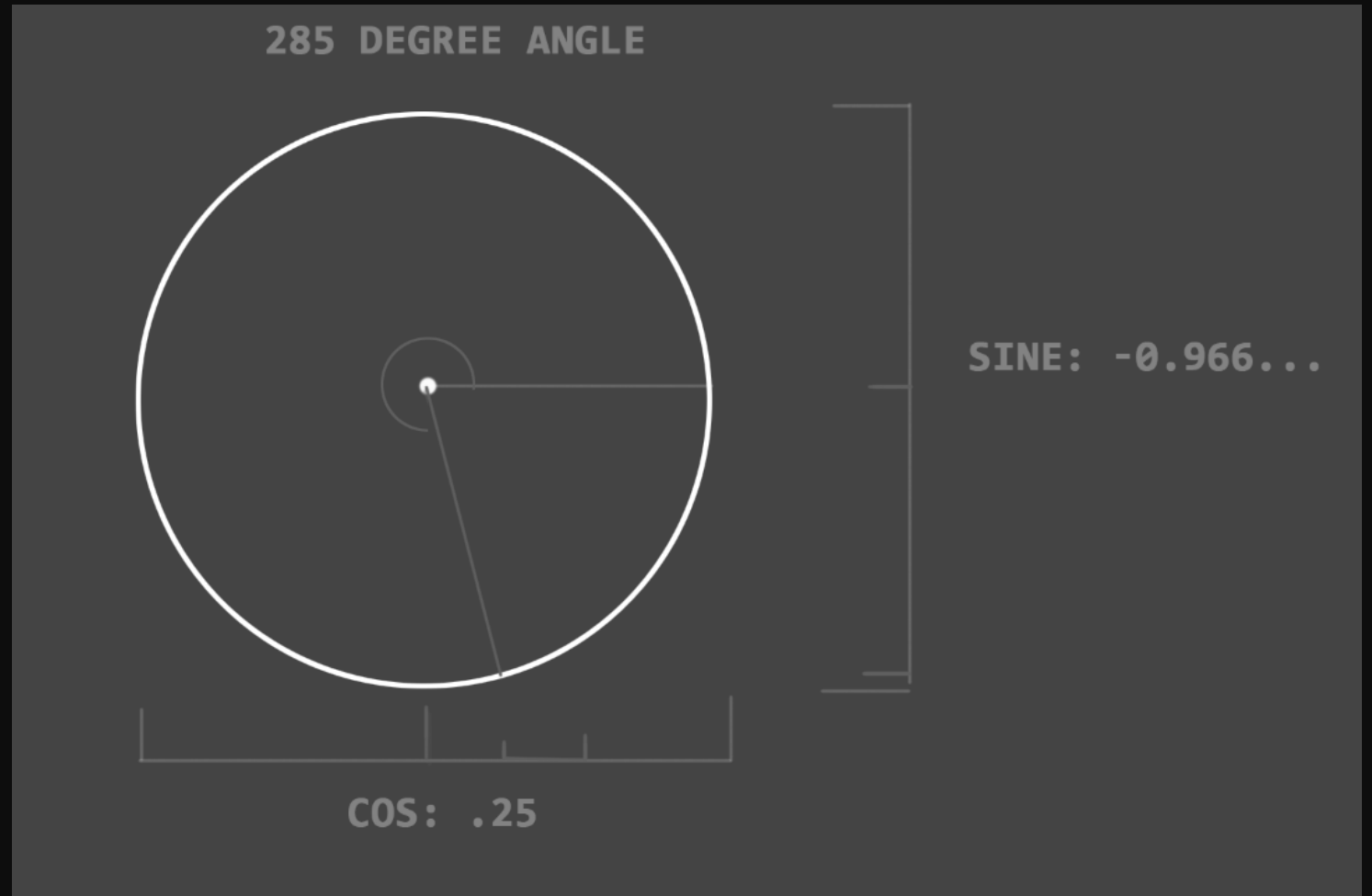
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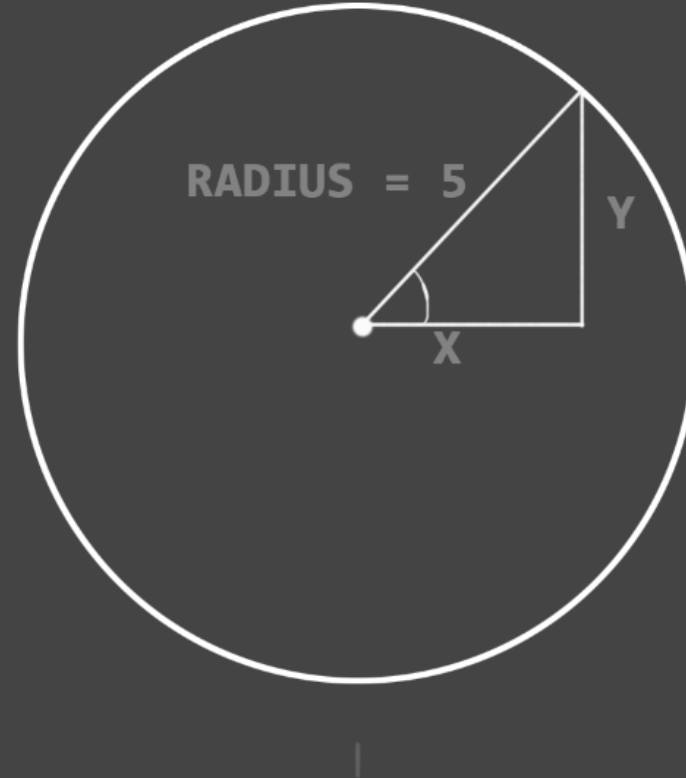
# UNIT CIRCLE

- Since this is a unit circle, with length 1 radius, the values are exactly at the sin and cos values.
- If the radius was larger, it would be multiplied by the sin and cos to get the x and y position of the point in the circle.



# UNIT CIRCLE

- In this example, we are seeing a larger radius, (which we are solving for using our standard Sine function )
- Sine = Opposite/Hypotenuse
- Since the Sine is giving us the proportion of the contribution to Y given the angle, we can use this to solve for the side of the triangle we don't know.



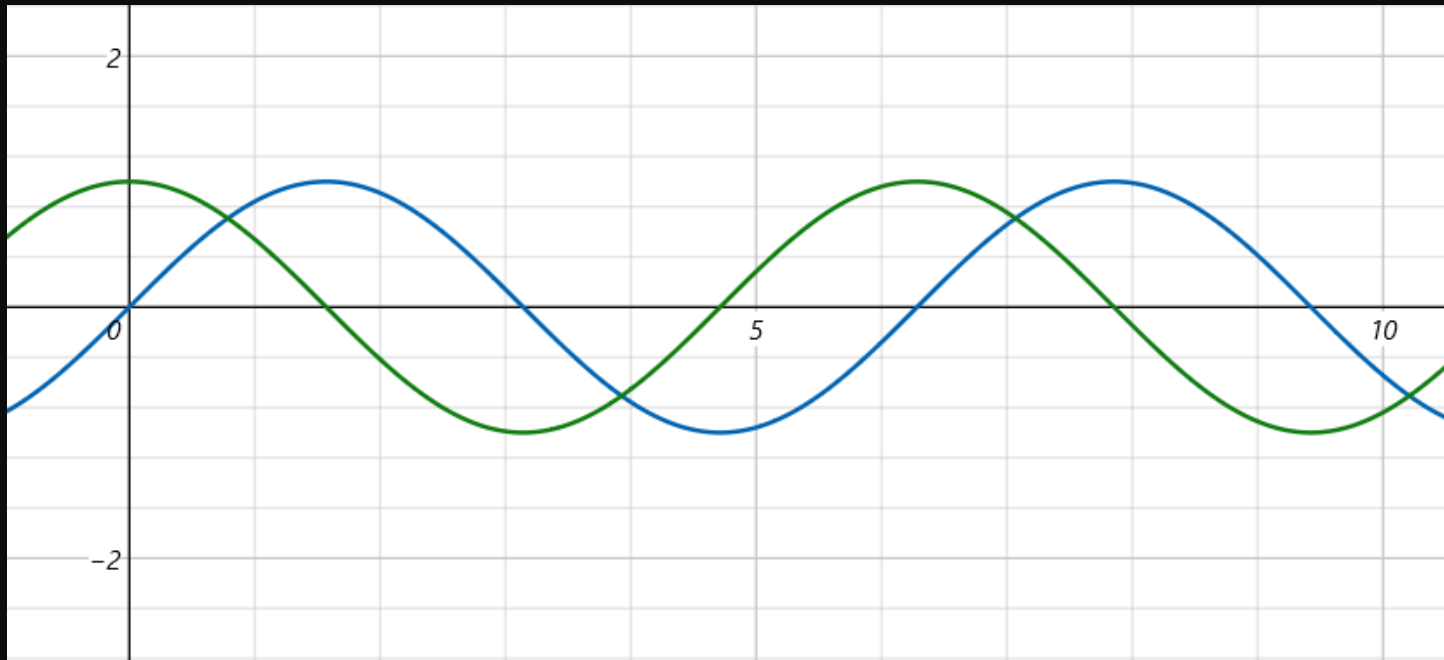
**SOH**

$$\begin{aligned}\text{SIN} &= \text{OPPOSITE} / \text{HYP} \\ \text{SIN}(45) &= 0.707 = Y/H \\ 0.707(5) &= Y\end{aligned}$$

# UNIT CIRCLE

- In this example, we are seeing a larger radius, (which we are solving for using our standard Sine function )
- $\text{Sine} = \text{Opposite} / \text{Hypotenuse}$
- Since the Sine is giving us the proportion of the contribution to Y given the angle, we can use this to solve for the side of the triangle we don't know.

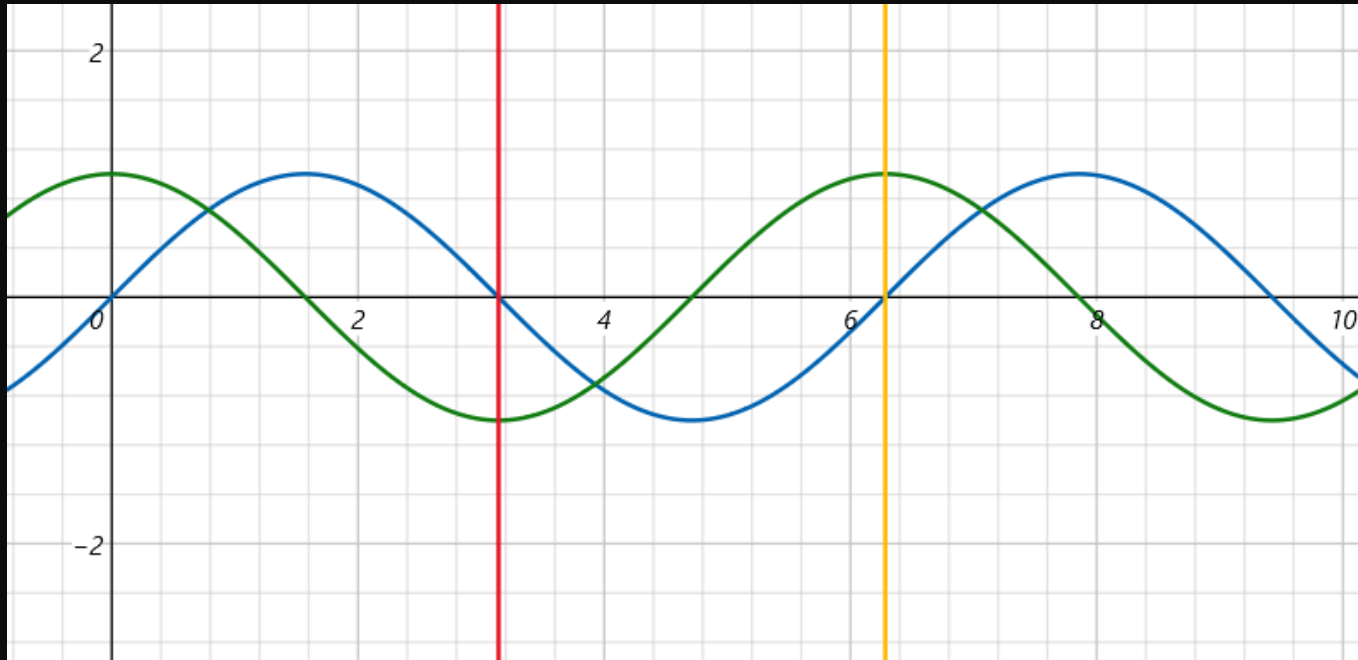
# SINE AND COSINE WAVE FUNCTIONS



|       |               |
|-------|---------------|
| $f_1$ | $y = \sin(x)$ |
| $f_2$ | $y = \cos(x)$ |

- If we plot the Sin and Cos functions over X on a graph, we end up with a Sine Wave and Cosine Wave functions. These are particularly useful in game development for creating smoothly oscillating curves of all types.
- What is really happening here though?

# SINE AND COSINE WAVE FUNCTIONS



|                                                                                     |               |
|-------------------------------------------------------------------------------------|---------------|
|  | $y = \sin(x)$ |
| $f_2$                                                                               | $y = \cos(x)$ |
| $f_3$                                                                               | $x = \pi$     |
| $f_4$                                                                               | $x = 2\pi$    |

- By setting  $X = \pi$ , I am effectively marking the half rotation of the circle (Red) as well as a full rotation of the circle (Yellow). You can see that at those points, the sine wave intercepts the X axis, and the Cosine wave it at the values (1) and (-1).
- This is the value of  $\pi$  and  $2\pi$ , which are a half rotation of a circle and full rotation of a circle.